BGP Tracer: Two-Phase Joint Traffic-Routing Attribution

Phase 1: Event-Traffic Correlation · Phase 2: PNS-inspired Causal Attribution · Five-class Verdicts

Data Inputs

Clou… RIP…
H…
5-m…

RIPE Atlas probes
Active reachability
traceroute

Coarse Screen

Baseline Choice
periodic daily/weekly
else robust median+MAD
no AS excluded

Phase 1

4 Parallel Detectors
Traffic | Outage | Leak | Hijack
Suspension - Temporal Correlation
c in {real_incident, silent_alert, ...}
c in {real_incident, silent_alert, ...}

Shared Cache

Shared Cache
Structured payload + confidence
Address (a, e, t, k) for audit
Degraded-mode log

Phase 2

PNS-inspired + HCAF
6 Causal Evidence Channels
Bayesian Synthesis
Direction: RtoT, TtoR, or abstain
Abstention is a verdict

Output

Five-class
Verdicts +
LLM report
(renderer)

Phase 1 Zoom: Deterministic Arbitration

Traffic Detector
baseline residual
anomaly in Ta(t)

Outage Detector
Routing withdrawal
+ Atlas drop

Leak Detector
AS-path anomaly
prefix overlap

Hijack Detector
Origin conflict
vs ROA / RPKI

Suspension
Evidential Gating
c in {real_incident, silent_alert, ...}
c in {real_incident, silent_alert, ...}

All four detectors run in parallel. Traffic detector: 0/7 hijack victims (key negative discriminator).

Abstention floor τ=0.50 on the 30-event curated set (26 cached).

Phase 2 Zoom: PNS-inspired Causal Attribution

MSBD Breakpoint
τ* abstention boundary

PDM: Do-Calculus
CSS = E[Y|do(X=1)] - E[Y|do(X=0)]

CSS Bootstrap

Transfer Entropy

MSBD Breakpoint

Bayesian Synthesis
HCAF score (0–1)
Direction: RtoT, TtoR, or abstain
(high confidence to auto-escalate)

CSS alone: 15 of 26 committed, 2 wrong. Six channels: 2 committed, 2 wrong.

Abstention is a verdict. LLM is a narrative renderer, not a verdict source.

Output and NOC Loop

Phase-1 Verdicts (Correlation)
c in {real_incident, silent_alert, traffic_only}

Traffic Flag
f_T in {0,1}
CSS magnitude

Direction
f_D in {0,1,abstain}
RtoT or TtoR

Evidence Report
LR, CI, p-value, HCAF, confidence

LLM Narrative
(renderer only)
structured verdict

High-confidence verdicts escalate to the NOC.
The rest are flagged for human review.

Top-1 Accuracy
86.7% (30 curated events)

Alert Reduction
9.3:1 (130 adjudicated live alerts)

End-to-end Latency
225.6 s median per event

Causal Accuracy
HCAF: 2 of 26 committed, both wrong